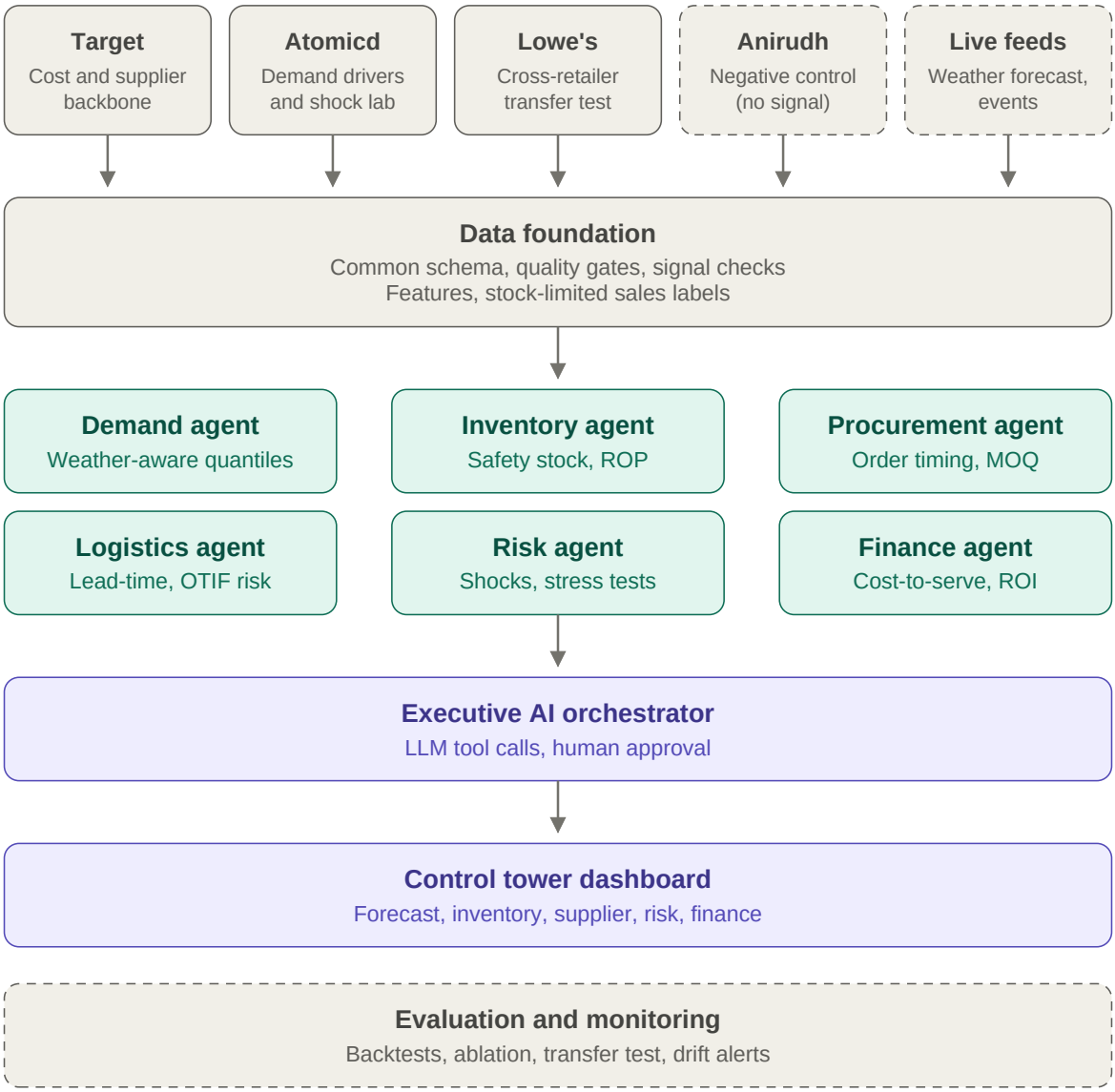
AI-Powered Control Tower: Derived Architecture

Derived from the Target, Atomicd, Lowe's and Anirudh retail datasets



Dashed boxes are control or production-time elements, not training data: Anirudh is a negative control, live feeds are added at deployment, and evaluation spans every layer.

What each dataset is used for

Each dataset was profiled before assigning it a role. The headline numbers below come from that profiling.

- **Target is the cost and supplier backbone.** 126,000 rows (70 SKUs x 10 stores x 180 days). It is the only file with holding, ordering, transport and stockout costs, supplier lead time and OTIF. Promo days sell about 30% more and weekends about 14% more. It contains no stockout events.
- **Atomicd is the demand-driver and shock lab.** 76,000 rows (20 products x 5 stores x 760 days, Jan 2022 to Jan 2024). It is the first file with a measurable external signal: sunny days sell about 20% more than snowy ones, promotion days about 26% more, and epidemic days about 36% less. Weather varies by store and day, and the epidemic flag is a network-wide regime on 152 of 760 days. In 8,654 rows (11.4%) units sold equal inventory, which gives a usable stock-limited-sales proxy. Caveats: each product appears under five different categories, so use Product ID, not Category; the Demand column is a noisy latent target (units sold exceeds it in 28% of rows); and there are no cost or supplier fields.
- **Lowe's is the cross-retailer transfer test.** 101,088 rows (72 SKUs x 12 stores x 117 days). Train on Target and test on Lowe's. Its fact-table category column is shuffled and must be re-joined from the product table, there is no supplier key, and it has no stockouts.
- **Anirudh is a negative control.** 73,100 of its keys (date, store, product) match Atomicd exactly, but the values are unrelated (units sold correlation 0.003), so it is a signal-free variant of the same schema and must not be merged with Atomicd. Adding weather, promo and price features changes its WAPE from 64.08% to 64.07%, which is no lift, and its forecast column leaks (0.997 correlation with actuals). Use it to prove the data-quality gate catches bad data.

Which dataset supports which agent

Agent	Target	Atomicd	Lowe's	Anirudh
Demand	Train: promo, weekday, price	Train: weather, promo, discount, epidemic, competitor price	Transfer test	Negative control
Inventory	Policy inputs: safety stock, ROP, costs	Inventory and order history	Transfer test	Not used
Procurement	MOQ, lead time, ordering cost	Units ordered only	MOQ, lead time; no supplier key	Not used
Logistics	Lead-time variability, OTIF by supplier	Not available	Anonymous lead time, OTIF	Not used
Risk	Simulator baseline (no stockouts)	Epidemic shock; stock-limited sales (11.4% of rows)	Not used	Not used
Finance	Holding, ordering, transport, stockout costs	No cost fields	Cost fields, partial	Not used

Evidence and design decisions

Quick, untuned backtests on a 28-day holdout (daily SKU-store WAPE, lower is better). They are indicative, not final model results.

Dataset	Seasonal naive	28-day moving avg	Model, history only	Model with drivers
Atomicd	47.0%	37.6%	34.1%	27.4%
Target	35.3%	26.2%	not run	23.5%
Lowe's	73.0%	53.4%	not run	52.2%
Anirudh	86.2%	64.7%	64.1%	64.1%

Atomicd shows that external drivers earn their place: weather, promo, discount, competitor price and the epidemic flag cut WAPE from 34.1% to 27.4%. Anirudh shows the opposite, which is why it works as a control.

Design decisions the data forced

1. **The data foundation has quality gates and a signal check.** It re-joins Lowe's categories, handles the missing supplier key, keys everything on Product ID because Atomicd and Anirudh categories are unreliable, flags forecast columns that correlate above 0.95 with actuals, and labels stock-limited sales in Atomicd.
2. **The Demand agent uses external drivers and quantile forecasts.** Driver effects are learned on Atomicd; retail patterns on Target. Caveat: Atomicd weather is observed on the day, so production needs a weather forecast feed, and the backtest lift is an upper bound.
3. **The Risk agent detects shocks and runs stress tests.** Atomicd's epidemic regime (about 36% lower demand) sets the shock size, and its 11.4% stock-limited rate is the calibration target. Target and Lowe's still have no stockouts, so simulation is still needed for their cost impact.
4. **The Logistics agent works at supplier level.** Only Target links lead time and OTIF to named suppliers; there is no shipment data anywhere.
5. **The Finance agent runs on Target's cost fields.** Atomicd has none, so any Atomicd-based scenario uses Target's cost parameters as a documented assumption.
6. **The orchestrator never computes numbers itself.** It reads the agents' outputs and recommends, and a human approves.
7. **Evaluation is built in.** Rolling-origin backtests on Atomicd (760 days allow several folds), driver on/off ablation, Target-to-Lowe's transfer, and Anirudh as a control that must show zero lift.

Restating the 75% to 90% accuracy KPI

At daily SKU-store level the honest results are about 73% accuracy on Atomicd and 76.5% on Target. Reaching 90% is unlikely because daily volumes are small and noisy. A defensible restatement is a WAPE reduction against the seasonal-naive baseline: about 42% on Atomicd (47.0% to 27.4%) and about 33% on Target (35.3% to 23.5%), with weekly and category-level accuracy reported alongside.

What changed from the previous version

Atomicd now leads the demand and risk story because it has real weather, promo and epidemic signal and a stockout proxy. Anirudh is demoted from sandbox to negative control after the overlap and zero-lift checks. Target remains the backbone for cost, supplier and logistics, and Lowe's remains the transfer test.